

MEDISATHI

Comprehensive Technical Architecture & MVP Video Presentation Guide

Project: MediSathi Mobile MVP	Target Event: CODEX Hackathon 2026
Tech Stack: Flutter, Java/Kotlin, OpenCV/MLKit, Android APIs	Target OS: Android (Offline-First)

1. Executive Summary & System Philosophy

MediSathi is a specialized, offline-first medical adherence and pill-verification ecosystem specifically designed for elderly patients and rural Indian healthcare settings. It resolves two critical healthcare challenges: **(1) Accidental consumption of lookalike pharmaceutical strips**, and **(2) Missed medication doses due to standard app notifications being ignored during phone sleep mode**.

The system combines a **Flutter-based elderly-accessible UI** with a **Native Java/Android Alarm Engine** operating directly at the Android OS layer. This guarantees 100% reliable full-screen lockscreen alarm triggering even when the app process is terminated or the device is in deep sleep mode.

2. Full Stack Technology Overview

Layer	Technologies Used	Core Responsibility
Frontend Shell	Flutter v3.x, Dart 3, Material 3 Design	Elderly-friendly responsive UI, 30pt+ typography, HSL green contrast theme, multi-language (EN/HI/MR).
State & Routing	Flutter Riverpod, GoRouter	Reactive state management, compile-safe dependency injection, declarative navigation back-stack.
Native Backend	Java, Kotlin, Android SDK (API 24-34)	Native AlarmActivity, AlarmReceiver, AlarmManager.setAlarmClock(), PowerManager.FULL_WAKE_LOCK.
Inter-Process Bridge	Android MethodChannel (com.bugbusters.../alarm)	Asynchronous bidirectional communication bridge connecting Dart logic with Native Android OS APIs.
Computer Vision	Google ML Kit OCR, Camera Stream API	On-device offline OCR text recognition, pill strip active ingredient extraction, blur & contrast validation.
Data Persistence	SharedPreferences v2, Offline CSV/JSON Catalog	Persistent storage for saved medicines, daily dose logs timeline, interaction rules database, and caregiver logs.
Telephony Integration	Android Intent Scheme (tel: & sms:)	Direct phone dialing and automated emergency SMS escalation to designated family caregivers.

3. Deep-Dive: Native Lock-Screen Alarm System

Standard Flutter notifications are frequently suppressed by Android 10+ battery optimizations or ignored when the phone is locked. MediSathi solves this by bypassing Flutter's UI thread for alarm execution using a multi-layer native Android architecture:

A. Hardware Screen Wake & Lock-Screen Bypass (Java/WindowManager)

The native `AlarmActivity.java` is configured with explicit window flags to take over the phone screen:

- `setShowWhenLocked(true)` & `setTurnScreenOn(true)`: Forces device display to turn ON immediately.
- `FLAG_KEEP_SCREEN_ON` & `FLAG_DISMISS_KEYGUARD`: Unlocks keyguard for the alarm dialog.
- `PowerManager.FULL_WAKE_LOCK | ACQUIRE_CAUSES_WAKEUP`: Wakes device CPU from deep sleep (Doze mode).

B. System Alarm Priority via `AlarmManager.setAlarmClock()`

In `MainActivity.kt`, alarms are scheduled using `AlarmManager.setAlarmClock()` instead of standard background intents. In Android OS, `setAlarmClock` grants high-priority system privileges, displaying a system alarm icon and ensuring execution regardless of battery saver settings.

C. Dual Execution Flow (Locked vs. Active State)

- **When Phone is Locked / Screen Off**: `AlarmReceiver.java` detects lock state (`isKeyguardLocked`) and directly launches `AlarmActivity` full-screen over the lockscreen with looping ringtone.
- **When Phone is Unlocked / Active in another app**: Displays a high-priority **Heads-Up Banner Notification** at the top of the screen to prevent disruptive app switching.

4. Computer Vision & Verification Pipeline

The pill strip verification engine operates 100% offline to ensure safety in remote areas without internet access. It consists of four deterministic verification gates:

Verification Gate	Mechanism	Safety Outcome
1. Quality Gate	Frame blur detection & lighting contrast analysis.	Rejects blurry or unreadable scans before OCR execution.
2. OCR Text Engine	Google ML Kit text recognition extracts brand/canonical names.	Parses pharmaceutical text from physical blister pack.
3. Shade & Lookalike Gate	Cross-references strip color signature against database lookalike groups.	Flags lookalike packaging confusion (e.g. Paracetamol vs. Amlodipine).
4. Confidence Gate	Multi-factor scoring rules (0.0 to 1.0 confidence).	Enforces high threshold before marking medicine safe to consume.

5. Smart Adherence & Caregiver Escalation

Adherence tracking is synchronized between local user UI and caregiver notification channels:

- **Adherence Logging:** Every **TAKEN** or **MISSED** action is recorded with ISO-8601 timestamps and persisted in SharedPreferences.
- **Today's Medication Timeline:** Displays daily dose history with status badges.
- **Caregiver Escalation:** When a dose is marked MISSED or unconfirmed beyond grace window, a MISSED_DOSE event is automatically generated in the Caregiver Center, enabling one-tap phone calls and pre-populated SMS alerts.

6. Step-by-Step Script for MVP Video Demonstration

Use this structured scene-by-scene script while recording the application video walkthrough for hackathon evaluation:

Scene / Time	Visual Action on Screen	Key Script Narration Point
Scene 1 (0:00 - 0:15)	Show Home Dashboard. Highlight clean green theme, large buttons, and daily reminder cards.	"Welcome to MediSathi — an offline-first, elderly-friendly medication safety system built for CODEX 2026."
Scene 2 (0:15 - 0:45)	Tap Scan Strip . Aim camera at a pill strip. Perform verification scan. Show verification results screen.	"MediSathi's on-device AI runs quality, OCR, and lookalike shade checks offline to verify medicine identity and warn against drug interactions."
Scene 3 (0:45 - 1:15)	Go to Reminders . Tap Set Reminder . Add medicine for 1 minute ahead. Save reminder.	"Setting a reminder configures a high-priority exact alarm natively using Android's AlarmManager."
Scene 4 (1:15 - 1:45)	Lock phone screen completely / turn off display . Wait for alarm time.	"Watch as the phone screen automatically wakes UP from deep sleep, bypasses the lockscreen, and displays the full-screen Medicine Time alarm!"

Scene 5 (1:45 - 2:15)	Tap ✓ Mark as Taken on alarm UI. Open app -> show Today's Adherence Timeline .	<i>"Marking taken logs adherence instantly. If a dose is missed, MediSathi automatically alerts designated family caregivers via SMS."</i>
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